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**Editor-in-Chief and Editorial Board**

*SEMA Journal*

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Dear Editors,

I am writing to submit our manuscript entitled "**Galois Invariants in Cyclotomic Lattice Enumeration: Deterministic Collapse of the SVP Search Space via Modular Projection Pruning**" for consideration for publication in *SEMA Journal*.

Recognizing the journal's commitment to advancing applied mathematics and scientific computing, we hope this work offers a meaningful contribution to the computational analysis of lattice-based structures, a field of growing relevance in modern cryptography.

Our study focuses on the Shortest Vector Problem (SVP) within cyclotomic lattices, which underpins the security of current Post-Quantum Cryptography (PQC) standards such as ML-KEM. While existing security models generally assume that Learning With Errors (LWE) noise is uniformly distributed across the phase space, our research aims to explore the inherent arithmetic constraints of these algebraic structures.

To this end, we introduce a "Galois Pruning" mechanism and evaluate it using a parallel Meet-in-the-Middle (MitM) computational architecture developed in C++. Our numerical simulations provide strong evidence that the modular projections of the secret vector act as deterministic algebraic filters. As detailed in the manuscript, this approach yields a measurable collapse of the search space—proportional to the product of the oracle primes ( $\prod p_i$ )—achieving a significant node elision in tested dimensions up to  $d = 32$ .

We believe these findings respectfully offer a new perspective on the uniform ergodicity assumption of discrete noise in structured lattices. While fully acknowledging the robust encapsulation mechanisms of current NIST standards, our empirical results suggest that the asymptotic complexity of SVP may benefit from further structural algebraic cryptanalysis.

To ensure complete transparency and support the reproducibility of our methodology, the theoretical framework, numerical data, and the optimized source code have been made publicly available via a permanent Zenodo repository (<https://doi.org/10.5281/zenodo.19920083>).

I confirm that this manuscript represents original work, has not been published elsewhere, and is not currently under consideration by any other journal.

Thank you for your time, consideration, and the vital work you do for the applied mathematics community. We would be grateful for the opportunity to receive feedback from your esteemed peer reviewers.

Sincerely,